

ABC RETAIL LTD ENTERPRISE DATA MODEL

Document Control

Item	Details
Document Title	ABC Retail Ltd Enterprise Data Model
Document Code	DABA-ABC-001
Version	1.0
Effective Date	To be approved
Review Frequency	Annually
Document Owner	Academic Affairs and Technology Department
Approval Authority	Academy President
Classification	Internal Academic Simulation Document

Version History

Version	Date	Description	Author	Approver
1.0	To be confirmed	Initial enterprise data model	Academic Affairs and Technology	Academy President

1. Introduction

ABC Retail Ltd is the primary enterprise simulation used by Data Analysis Blueprint Academy.

It is designed to provide learners with a realistic business environment in which they can develop practical skills in:

- Microsoft Excel.
- SQL.
- Power BI.
- Business statistics.
- Data cleaning.
- Data visualization.
- Business intelligence.
- Artificial intelligence.
- Data governance.
- Information security.
- Financial analysis.
- Human resources analytics.
- Sales analytics.

- Inventory analytics.
- Procurement analytics.
- Customer analytics.
- Capstone project development.

Rather than using unrelated datasets for every lesson, DABA students will work with connected datasets from one simulated company.

This allows students to understand how departments, systems, transactions, and business decisions are connected.

2. Enterprise Profile

2.1 Company Name

ABC Retail Ltd

2.2 Industry

Retail and Distribution

2.3 Headquarters

Douala, Cameroon

2.4 Business Model

ABC Retail Ltd purchases products from suppliers and distributes them through retail branches, wholesale channels, online sales, and corporate contracts.

2.5 Organizational Size

Category	Estimated Size
Employees	500
Branches	30
Warehouses	5
Products	300+
Suppliers	100+
Customers	20,000+
Annual Transactions	100,000+
Departments	12

Category	Estimated Size
Regions	5

2.6 Operational Regions

ABC Retail Ltd operates across:

- Littoral.
- Centre.
- South West.
- North West.
- West.

Future expansion may include other regions.

3. Purpose of the Enterprise Simulation

The ABC Retail Ltd simulation will help learners:

- Understand business operations.
 - Work with relational data.
 - Clean realistic datasets.
 - Identify data quality problems.
 - Build formulas and calculations.
 - Write SQL queries.
 - Develop dashboards.
 - Analyze departmental performance.
 - Detect trends and anomalies.
 - Prepare management reports.
 - Practice responsible AI.
 - Build professional portfolios.
 - Complete capstone projects.
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4. Main Business Departments

ABC Retail Ltd includes the following departments:

1. Executive Management.
2. Finance and Accounting.
3. Human Resources.
4. Sales.
5. Marketing.
6. Procurement.
7. Inventory and Warehousing.
8. Logistics and Distribution.

9. Customer Service.
10. Information Technology.
11. Information Security.
12. Business Intelligence and Analytics.

Each department will produce and consume business data.

5. Enterprise Data Architecture

The enterprise data model will be organized into five major layers.

5.1 Master Data

Master data represents relatively stable business entities.

Examples:

- Employees.
- Customers.
- Suppliers.
- Products.
- Branches.
- Departments.
- Warehouses.
- Accounts.
- Regions.
- Job positions.

5.2 Transaction Data

Transaction data records business activities.

Examples:

- Sales.
- Payments.
- Purchases.
- Stock movements.
- Payroll.
- Expenses.
- Deliveries.
- Returns.
- Customer complaints.
- Attendance.

5.3 Reference Data

Reference data supports classification and consistency.

Examples:

- Product categories.
- Payment methods.
- Employment status.
- Customer type.
- Transaction status.
- Complaint category.
- Leave type.
- Risk level.
- Region code.
- Currency.

5.4 Analytical Data

Analytical data includes calculated and summarized information.

Examples:

- Monthly revenue.
- Branch performance.
- Employee turnover.
- Inventory turnover.
- Customer lifetime value.
- Sales targets.
- Profit margins.
- Supplier performance.
- Marketing conversion.
- Risk indicators.

5.5 Audit and Governance Data

Audit data supports control, security, and traceability.

Examples:

- Login activity.
- Data changes.
- User access.
- Approval history.
- System incidents.
- Audit findings.
- Risk treatment.
- Policy acknowledgements.

6. Core Database Domains

The full enterprise database will be divided into the following domains.

Domain Code	Domain Name
COR	Corporate Structure
HR	Human Resources
CRM	Customer Relationship Management
SAL	Sales
PRD	Product Management
INV	Inventory
PUR	Procurement
LOG	Logistics
FIN	Finance
MKT	Marketing
CSR	Customer Service
IT	Information Technology
SEC	Information Security
BI	Business Intelligence
GOV	Governance and Audit

7. Corporate Structure Tables

7.1 Regions

Table Name: regions

Field	Data Type	Description
region_id	Integer	Unique region identifier
region_code	Varchar	Short region code
region_name	Varchar	Region name
country	Varchar	Country
status	Varchar	Active or inactive
created_at	Datetime	Record creation date

7.2 Branches

Table Name: branches

Field	Data Type	Description
branch_id	Integer	Unique branch identifier
branch_code	Varchar	Branch code
branch_name	Varchar	Branch name
region_id	Integer	Linked region
branch_type	Varchar	Retail, warehouse, office, or hybrid
address	Varchar	Branch address
city	Varchar	City
opening_date	Date	Opening date
manager_employee_id	Integer	Branch manager
phone	Varchar	Branch contact
email	Varchar	Branch email
status	Varchar	Branch status
created_at	Datetime	Record creation date

7.3 Departments

Table Name: departments

Field	Data Type	Description
department_id	Integer	Unique department identifier
department_code	Varchar	Department code
department_name	Varchar	Department name
manager_employee_id	Integer	Department manager
annual_budget	Decimal	Approved annual budget
location	Varchar	Primary location
status	Varchar	Department status
created_at	Datetime	Record creation date

7.4 Job Positions

Table Name: job_positions

Field	Data Type	Description
position_id	Integer	Unique position identifier

Field	Data Type	Description
position_code	Varchar	Position code
position_title	Varchar	Job title
department_id	Integer	Department
job_level	Varchar	Entry, junior, mid, senior, management
minimum_salary	Decimal	Minimum salary
maximum_salary	Decimal	Maximum salary
employment_type	Varchar	Permanent, temporary, contract, intern
status	Varchar	Position status

8. Human Resources Tables

8.1 Employees

Table Name: employees

Field	Data Type	Description
employee_id	Integer	Unique employee identifier
employee_number	Varchar	Official employee number
first_name	Varchar	First name
last_name	Varchar	Last name
gender	Varchar	Gender
date_of_birth	Date	Date of birth
email	Varchar	Work email
phone	Varchar	Phone number
branch_id	Integer	Assigned branch
department_id	Integer	Assigned department
position_id	Integer	Job position
manager_id	Integer	Reporting manager
hire_date	Date	Employment start date
employment_type	Varchar	Employment category
employment_status	Varchar	Active, suspended, resigned, terminated
base_salary	Decimal	Monthly base salary

Field	Data Type	Description
bank_name	Varchar	Employee bank
account_number_masked	Varchar	Masked account identifier
created_at	Datetime	Record creation date
updated_at	Datetime	Last update date

8.2 Employee Attendance

Table Name: employee_attendance

Field	Data Type	Description
attendance_id	Integer	Unique attendance record
employee_id	Integer	Employee
attendance_date	Date	Date
clock_in	Time	Arrival time
clock_out	Time	Departure time
attendance_status	Varchar	Present, absent, late, leave
hours_worked	Decimal	Total hours worked
overtime_hours	Decimal	Overtime
branch_id	Integer	Work location
remarks	Varchar	Attendance remarks

8.3 Employee Leave

Table Name: employee_leave

Field	Data Type	Description
leave_id	Integer	Unique leave request
employee_id	Integer	Employee
leave_type_id	Integer	Leave category
start_date	Date	Leave start
end_date	Date	Leave end
days_requested	Decimal	Number of days
reason	Text	Leave reason
approval_status	Varchar	Pending, approved, rejected

Field	Data Type	Description
approved_by	Integer	Approving manager
approval_date	Date	Approval date
created_at	Datetime	Request date

8.4 Payroll

Table Name: payroll

Field	Data Type	Description
payroll_id	Integer	Unique payroll record
employee_id	Integer	Employee
payroll_month	Integer	Payroll month
payroll_year	Integer	Payroll year
base_salary	Decimal	Base salary
allowances	Decimal	Total allowances
overtime_pay	Decimal	Overtime payment
bonuses	Decimal	Bonuses
deductions	Decimal	Total deductions
tax	Decimal	Payroll tax
net_salary	Decimal	Net salary
payment_date	Date	Payment date
payment_status	Varchar	Pending, paid, failed
approved_by	Integer	Approving officer

8.5 Employee Performance

Table Name: employee_performance

Field	Data Type	Description
performance_id	Integer	Unique performance review
employee_id	Integer	Employee
review_period	Varchar	Review period
reviewer_id	Integer	Reviewer
productivity_score	Decimal	Productivity rating

Field	Data Type	Description
quality_score	Decimal	Quality rating
attendance_score	Decimal	Attendance rating
teamwork_score	Decimal	Teamwork rating
overall_score	Decimal	Overall rating
performance_level	Varchar	Poor, fair, good, excellent
review_date	Date	Review date
comments	Text	Reviewer comments

8.6 Employee Training

Table Name: employee_training

Field	Data Type	Description
training_record_id	Integer	Unique training record
employee_id	Integer	Employee
training_course_id	Integer	Training course
start_date	Date	Start date
completion_date	Date	Completion date
provider	Varchar	Training provider
result	Varchar	Completed, failed, ongoing
score	Decimal	Final score
certificate_number	Varchar	Certificate reference
cost	Decimal	Training cost

9. Customer Tables

9.1 Customers

Table Name: customers

Field	Data Type	Description
customer_id	Integer	Unique customer identifier
customer_code	Varchar	Customer code

Field	Data Type	Description
first_name	Varchar	First name
last_name	Varchar	Last name
company_name	Varchar	Company name if applicable
customer_type	Varchar	Individual, retail, wholesale, corporate
gender	Varchar	Gender where collected
date_of_birth	Date	Date of birth where collected
email	Varchar	Email
phone	Varchar	Phone
city	Varchar	City
region_id	Integer	Region
registration_date	Date	Date registered
loyalty_status	Varchar	Standard, silver, gold, platinum
credit_limit	Decimal	Approved credit
customer_status	Varchar	Active, inactive, blocked
preferred_channel	Varchar	Store, phone, email, online
created_at	Datetime	Record creation date

9.2 Customer Addresses

Table Name: customer_addresses

Field	Data Type	Description
address_id	Integer	Unique address
customer_id	Integer	Customer
address_type	Varchar	Billing, shipping, home, office
street_address	Varchar	Street
city	Varchar	City
region_id	Integer	Region
country	Varchar	Country
is_primary	Boolean	Primary address flag

9.3 Customer Loyalty Transactions

Table Name: loyalty_transactions

Field	Data Type	Description
loyalty_transaction_id	Integer	Unique record
customer_id	Integer	Customer
transaction_date	Date	Date
transaction_type	Varchar	Earned, redeemed, expired, adjusted
points	Integer	Points
reference_number	Varchar	Linked transaction
balance_after	Integer	Remaining points

10. Product Management Tables

10.1 Product Categories

Table Name: product_categories

Field	Data Type	Description
category_id	Integer	Unique category
category_code	Varchar	Category code
category_name	Varchar	Category name
parent_category_id	Integer	Parent category
description	Text	Category description
status	Varchar	Active or inactive

10.2 Products

Table Name: products

Field	Data Type	Description
product_id	Integer	Unique product identifier
product_code	Varchar	Product code
product_name	Varchar	Product name
category_id	Integer	Product category
brand	Varchar	Brand
unit_of_measure	Varchar	Unit

Field	Data Type	Description
cost_price	Decimal	Purchase cost
selling_price	Decimal	Standard selling price
reorder_level	Integer	Minimum quantity
reorder_quantity	Integer	Recommended reorder quantity
supplier_id	Integer	Preferred supplier
tax_rate	Decimal	Tax percentage
product_status	Varchar	Active, inactive, discontinued
launch_date	Date	Product introduction date
expiry_tracking	Boolean	Whether expiry is tracked
created_at	Datetime	Creation date

10.3 Product Price History

Table Name: product_price_history

Field	Data Type	Description
price_history_id	Integer	Unique record
product_id	Integer	Product
old_price	Decimal	Previous price
new_price	Decimal	New price
effective_date	Date	Effective date
reason	Varchar	Price change reason
approved_by	Integer	Approving employee

11. Supplier and Procurement Tables

11.1 Suppliers

Table Name: suppliers

Field	Data Type	Description
supplier_id	Integer	Unique supplier
supplier_code	Varchar	Supplier code

Field	Data Type	Description
supplier_name	Varchar	Supplier name
contact_person	Varchar	Main contact
email	Varchar	Email
phone	Varchar	Phone
address	Varchar	Address
city	Varchar	City
country	Varchar	Country
payment_terms	Varchar	Payment terms
lead_time_days	Integer	Average delivery time
supplier_rating	Decimal	Performance rating
supplier_status	Varchar	Active, inactive, blocked
registration_date	Date	Date registered

11.2 Purchase Requisitions

Table Name: purchase_requisitions

Field	Data Type	Description
requisition_id	Integer	Unique requisition
requisition_number	Varchar	Requisition reference
department_id	Integer	Requesting department
requested_by	Integer	Requesting employee
request_date	Date	Request date
required_date	Date	Required delivery date
total_estimated_amount	Decimal	Estimated cost
approval_status	Varchar	Pending, approved, rejected
approved_by	Integer	Approver
approval_date	Date	Approval date
reason	Text	Procurement reason

11.3 Purchase Requisition Items

Table Name: purchase_requisition_items

Field	Data Type	Description
requisition_item_id	Integer	Unique item
requisition_id	Integer	Requisition
product_id	Integer	Product
quantity_requested	Integer	Requested quantity
estimated_unit_cost	Decimal	Estimated cost
estimated_total	Decimal	Estimated total

11.4 Purchase Orders

Table Name: purchase_orders

Field	Data Type	Description
purchase_order_id	Integer	Unique purchase order
purchase_order_number	Varchar	PO number
supplier_id	Integer	Supplier
order_date	Date	Order date
expected_delivery_date	Date	Expected delivery
branch_id	Integer	Receiving branch
subtotal	Decimal	Subtotal
tax_amount	Decimal	Tax
total_amount	Decimal	Total
currency	Varchar	Currency
order_status	Varchar	Draft, approved, sent, received, cancelled
created_by	Integer	Creator
approved_by	Integer	Approver

11.5 Purchase Order Items

Table Name: purchase_order_items

Field	Data Type	Description
purchase_order_item_id	Integer	Unique line item
purchase_order_id	Integer	Purchase order
product_id	Integer	Product

Field	Data Type	Description
quantity_ordered	Integer	Ordered quantity
unit_cost	Decimal	Cost per unit
discount	Decimal	Discount
tax_amount	Decimal	Tax
line_total	Decimal	Total line value
quantity_received	Integer	Received quantity

11.6 Supplier Performance

Table Name: supplier_performance

Field	Data Type	Description
supplier_performance_id	Integer	Unique record
supplier_id	Integer	Supplier
review_period	Varchar	Review period
delivery_score	Decimal	Delivery rating
quality_score	Decimal	Quality rating
price_score	Decimal	Price competitiveness
service_score	Decimal	Service rating
overall_score	Decimal	Overall rating
reviewed_by	Integer	Reviewer
review_date	Date	Review date

12. Inventory Tables

12.1 Warehouses

Table Name: warehouses

Field	Data Type	Description
warehouse_id	Integer	Unique warehouse
warehouse_code	Varchar	Warehouse code
warehouse_name	Varchar	Warehouse name

Field	Data Type	Description
branch_id	Integer	Linked branch
manager_employee_id	Integer	Warehouse manager
storage_capacity	Decimal	Total capacity
warehouse_status	Varchar	Active or inactive

12.2 Inventory Balance

Table Name: inventory_balance

Field	Data Type	Description
inventory_balance_id	Integer	Unique balance record
product_id	Integer	Product
warehouse_id	Integer	Warehouse
branch_id	Integer	Branch
quantity_on_hand	Integer	Current stock
quantity_reserved	Integer	Reserved stock
quantity_available	Integer	Available stock
reorder_level	Integer	Reorder threshold
last_stock_count_date	Date	Last physical count
updated_at	Datetime	Last update

12.3 Inventory Transactions

Table Name: inventory_transactions

Field	Data Type	Description
inventory_transaction_id	Integer	Unique movement
product_id	Integer	Product
warehouse_id	Integer	Warehouse
branch_id	Integer	Branch
transaction_date	Datetime	Date and time
transaction_type	Varchar	Receipt, sale, return, transfer, adjustment, damage
quantity	Integer	Quantity moved
unit_cost	Decimal	Unit cost

Field	Data Type	Description
reference_number	Varchar	Source transaction
created_by	Integer	Employee
remarks	Varchar	Notes

12.4 Stock Counts

Table Name: stock_counts

Field	Data Type	Description
stock_count_id	Integer	Unique count
warehouse_id	Integer	Warehouse
count_date	Date	Count date
conducted_by	Integer	Employee
approved_by	Integer	Approver
count_status	Varchar	Planned, in progress, completed
total_variance_value	Decimal	Value of differences

12.5 Stock Count Items

Table Name: stock_count_items

Field	Data Type	Description
stock_count_item_id	Integer	Unique item
stock_count_id	Integer	Stock count
product_id	Integer	Product
system_quantity	Integer	System balance
counted_quantity	Integer	Physical count
variance_quantity	Integer	Difference
variance_value	Decimal	Financial difference
reason	Varchar	Reason for variance

12.6 Product Batches

Table Name: product_batches

Field	Data Type	Description
batch_id	Integer	Unique batch
product_id	Integer	Product
batch_number	Varchar	Batch reference
manufacture_date	Date	Manufacture date
expiry_date	Date	Expiry date
quantity_received	Integer	Received quantity
quantity_remaining	Integer	Remaining quantity
warehouse_id	Integer	Warehouse
supplier_id	Integer	Supplier

13. Sales Tables

13.1 Sales Orders

Table Name: sales_orders

Field	Data Type	Description
sales_order_id	Integer	Unique sales order
order_number	Varchar	Order number
customer_id	Integer	Customer
branch_id	Integer	Branch
sales_employee_id	Integer	Sales representative
order_date	Datetime	Order date
sales_channel	Varchar	Store, online, wholesale, corporate
subtotal	Decimal	Subtotal
discount_amount	Decimal	Discount
tax_amount	Decimal	Tax
total_amount	Decimal	Total
payment_status	Varchar	Unpaid, partial, paid, refunded
order_status	Varchar	Pending, completed, cancelled, returned
delivery_required	Boolean	Delivery flag
created_at	Datetime	Created date

13.2 Sales Order Items

Table Name: sales_order_items

Field	Data Type	Description
sales_order_item_id	Integer	Unique line
sales_order_id	Integer	Sales order
product_id	Integer	Product
quantity	Integer	Quantity
unit_price	Decimal	Selling price
unit_cost	Decimal	Cost
discount	Decimal	Discount
tax_amount	Decimal	Tax
line_total	Decimal	Total
gross_profit	Decimal	Gross profit

13.3 Sales Targets

Table Name: sales_targets

Field	Data Type	Description
sales_target_id	Integer	Unique target
employee_id	Integer	Sales employee
branch_id	Integer	Branch
target_month	Integer	Month
target_year	Integer	Year
revenue_target	Decimal	Sales target
customer_target	Integer	New customer target
product_target	Integer	Product unit target
approved_by	Integer	Approver

13.4 Sales Returns

Table Name: sales_returns

Field	Data Type	Description
return_id	Integer	Unique return
return_number	Varchar	Return reference
sales_order_id	Integer	Original order
customer_id	Integer	Customer
branch_id	Integer	Branch
return_date	Date	Return date
return_reason	Varchar	Reason
total_refund_amount	Decimal	Refund total
return_status	Varchar	Pending, approved, rejected, completed
approved_by	Integer	Approver

13.5 Sales Return Items

Table Name: sales_return_items

Field	Data Type	Description
return_item_id	Integer	Unique return item
return_id	Integer	Return
product_id	Integer	Product
quantity_returned	Integer	Returned quantity
unit_price	Decimal	Unit price
refund_amount	Decimal	Refund value
condition	Varchar	Resalable, damaged, expired

14. Payment Tables

14.1 Customer Payments

Table Name: customer_payments

Field	Data Type	Description
payment_id	Integer	Unique payment
payment_number	Varchar	Payment reference

Field	Data Type	Description
customer_id	Integer	Customer
sales_order_id	Integer	Related order
payment_date	Datetime	Payment date
payment_method	Varchar	Cash, bank, card, mobile money, credit
amount	Decimal	Amount paid
currency	Varchar	Currency
transaction_reference	Varchar	External reference
payment_status	Varchar	Pending, completed, failed, reversed
received_by	Integer	Employee
branch_id	Integer	Branch

14.2 Refunds

Table Name: customer_refunds

Field	Data Type	Description
refund_id	Integer	Unique refund
payment_id	Integer	Original payment
return_id	Integer	Related return
refund_date	Date	Refund date
refund_amount	Decimal	Refund amount
refund_method	Varchar	Cash, bank, card, mobile money
refund_status	Varchar	Pending, approved, completed, rejected
approved_by	Integer	Approver
reason	Text	Refund reason

15. Marketing Tables

15.1 Marketing Campaigns

Table Name: marketing_campaigns

Field	Data Type	Description
campaign_id	Integer	Unique campaign
campaign_code	Varchar	Campaign code
campaign_name	Varchar	Campaign name
channel	Varchar	Email, social media, radio, event, SMS
start_date	Date	Start date
end_date	Date	End date
budget	Decimal	Campaign budget
target_audience	Varchar	Intended audience
campaign_status	Varchar	Planned, active, completed, cancelled
campaign_manager_id	Integer	Responsible employee

15.2 Marketing Leads

Table Name: marketing_leads

Field	Data Type	Description
lead_id	Integer	Unique lead
campaign_id	Integer	Campaign
lead_name	Varchar	Lead name
email	Varchar	Email
phone	Varchar	Phone
source	Varchar	Lead source
lead_date	Date	Date captured
lead_status	Varchar	New, contacted, qualified, converted, lost
assigned_to	Integer	Assigned employee
converted_customer_id	Integer	Customer if converted

15.3 Marketing Expenses

Table Name: marketing_expenses

Field	Data Type	Description
marketing_expense_id	Integer	Unique expense
campaign_id	Integer	Campaign

Field	Data Type	Description
expense_date	Date	Expense date
expense_type	Varchar	Advertising, event, printing, influencer
supplier_id	Integer	Supplier
amount	Decimal	Amount
approved_by	Integer	Approver
payment_status	Varchar	Pending or paid

16. Logistics Tables

16.1 Deliveries

Table Name: deliveries

Field	Data Type	Description
delivery_id	Integer	Unique delivery
delivery_number	Varchar	Delivery number
sales_order_id	Integer	Sales order
customer_id	Integer	Customer
delivery_address_id	Integer	Delivery address
dispatch_date	Datetime	Dispatch date
expected_delivery_date	Date	Expected date
actual_delivery_date	Date	Actual date
delivery_status	Varchar	Pending, dispatched, delivered, failed, returned
vehicle_id	Integer	Delivery vehicle
driver_employee_id	Integer	Driver
delivery_cost	Decimal	Delivery cost

16.2 Vehicles

Table Name: vehicles

Field	Data Type	Description
vehicle_id	Integer	Unique vehicle

Field	Data Type	Description
registration_number	Varchar	Vehicle registration
vehicle_type	Varchar	Truck, van, motorcycle
make	Varchar	Manufacturer
model	Varchar	Model
year	Integer	Manufacturing year
branch_id	Integer	Assigned branch
purchase_date	Date	Purchase date
vehicle_status	Varchar	Active, maintenance, inactive
current_mileage	Decimal	Current mileage

16.3 Vehicle Maintenance

Table Name: vehicle_maintenance

Field	Data Type	Description
maintenance_id	Integer	Unique maintenance record
vehicle_id	Integer	Vehicle
maintenance_date	Date	Date
maintenance_type	Varchar	Service, repair, inspection
description	Text	Work completed
supplier_id	Integer	Service provider
cost	Decimal	Maintenance cost
mileage	Decimal	Vehicle mileage
next_service_date	Date	Next planned service

17. Customer Service Tables

17.1 Customer Complaints

Table Name: customer_complaints

Field	Data Type	Description
complaint_id	Integer	Unique complaint

Field	Data Type	Description
complaint_number	Varchar	Complaint reference
customer_id	Integer	Customer
sales_order_id	Integer	Related order
complaint_date	Datetime	Date received
complaint_channel	Varchar	Phone, email, store, website, social media
complaint_category	Varchar	Product, delivery, service, payment, staff
complaint_description	Text	Complaint details
priority	Varchar	Low, medium, high, critical
assigned_to	Integer	Responsible employee
complaint_status	Varchar	Open, in progress, resolved, closed
resolution_date	Date	Date resolved
customer_satisfaction_score	Decimal	Satisfaction rating

17.2 Customer Interactions

Table Name: customer_interactions

Field	Data Type	Description
interaction_id	Integer	Unique interaction
customer_id	Integer	Customer
interaction_date	Datetime	Interaction date
channel	Varchar	Phone, email, chat, store
interaction_type	Varchar	Inquiry, complaint, follow-up, feedback
employee_id	Integer	Handling employee
interaction_summary	Text	Summary
next_action	Varchar	Required follow-up
follow_up_date	Date	Follow-up date

18. Finance Tables

18.1 Chart of Accounts

Table Name: chart_of_accounts

Field	Data Type	Description
account_id	Integer	Unique account
account_code	Varchar	Account code
account_name	Varchar	Account name
account_type	Varchar	Asset, liability, equity, revenue, expense
parent_account_id	Integer	Parent account
account_status	Varchar	Active or inactive

18.2 Journal Entries

Table Name: journal_entries

Field	Data Type	Description
journal_entry_id	Integer	Unique journal entry
journal_number	Varchar	Journal reference
transaction_date	Date	Transaction date
description	Text	Description
source_module	Varchar	Sales, purchases, payroll, inventory
reference_number	Varchar	Source reference
total_debit	Decimal	Total debit
total_credit	Decimal	Total credit
posting_status	Varchar	Draft, posted, reversed
created_by	Integer	Creator
approved_by	Integer	Approver

18.3 Journal Entry Lines

Table Name: journal_entry_lines

Field	Data Type	Description
journal_line_id	Integer	Unique line
journal_entry_id	Integer	Journal entry
account_id	Integer	Account
branch_id	Integer	Branch
department_id	Integer	Department

Field	Data Type	Description
debit_amount	Decimal	Debit
credit_amount	Decimal	Credit
description	Varchar	Line description

18.4 Expenses

Table Name: expenses

Field	Data Type	Description
expense_id	Integer	Unique expense
expense_number	Varchar	Expense reference
expense_date	Date	Expense date
department_id	Integer	Department
branch_id	Integer	Branch
expense_category	Varchar	Expense type
supplier_id	Integer	Supplier
description	Text	Expense description
amount	Decimal	Expense amount
tax_amount	Decimal	Tax
payment_status	Varchar	Pending, approved, paid
requested_by	Integer	Requester
approved_by	Integer	Approver

18.5 Budgets

Table Name: budgets

Field	Data Type	Description
budget_id	Integer	Unique budget
budget_year	Integer	Year
department_id	Integer	Department
branch_id	Integer	Branch
account_id	Integer	Account
budget_month	Integer	Month

Field	Data Type	Description
budget_amount	Decimal	Approved budget
revised_amount	Decimal	Revised budget
approved_by	Integer	Approver

18.6 Accounts Receivable

Table Name: accounts_receivable

Field	Data Type	Description
receivable_id	Integer	Unique receivable
customer_id	Integer	Customer
sales_order_id	Integer	Sales order
invoice_number	Varchar	Invoice reference
invoice_date	Date	Invoice date
due_date	Date	Payment due date
invoice_amount	Decimal	Invoice total
amount_paid	Decimal	Amount paid
outstanding_amount	Decimal	Balance
aging_category	Varchar	Current, 30, 60, 90+ days
receivable_status	Varchar	Open, partial, paid, overdue

18.7 Accounts Payable

Table Name: accounts_payable

Field	Data Type	Description
payable_id	Integer	Unique payable
supplier_id	Integer	Supplier
purchase_order_id	Integer	Purchase order
supplier_invoice_number	Varchar	Supplier invoice
invoice_date	Date	Invoice date
due_date	Date	Due date
invoice_amount	Decimal	Amount
amount_paid	Decimal	Amount paid

Field	Data Type	Description
outstanding_amount	Decimal	Balance
payable_status	Varchar	Open, partial, paid, overdue

19. Information Technology Tables

19.1 IT Assets

Table Name: it_assets

Field	Data Type	Description
asset_id	Integer	Unique asset
asset_tag	Varchar	Asset tag
asset_type	Varchar	Laptop, desktop, server, router, printer
manufacturer	Varchar	Manufacturer
model	Varchar	Model
serial_number	Varchar	Serial number
assigned_employee_id	Integer	Assigned employee
branch_id	Integer	Location
purchase_date	Date	Purchase date
purchase_cost	Decimal	Purchase cost
warranty_end_date	Date	Warranty end
asset_status	Varchar	Active, repair, lost, retired
last_inventory_date	Date	Last asset check

19.2 IT Support Tickets

Table Name: it_support_tickets

Field	Data Type	Description
ticket_id	Integer	Unique ticket
ticket_number	Varchar	Ticket reference
raised_by	Integer	Employee
branch_id	Integer	Branch

Field	Data Type	Description
created_at	Datetime	Date created
category	Varchar	Hardware, software, network, access
priority	Varchar	Low, medium, high, critical
description	Text	Issue details
assigned_to	Integer	Support employee
ticket_status	Varchar	Open, assigned, resolved, closed
resolved_at	Datetime	Resolution date
resolution_summary	Text	Resolution
satisfaction_score	Decimal	User rating

19.3 System Users

Table Name: system_users

Field	Data Type	Description
user_id	Integer	Unique system user
employee_id	Integer	Employee
username	Varchar	Login name
email	Varchar	Login email
role_id	Integer	Assigned role
account_status	Varchar	Active, disabled, locked
last_login	Datetime	Last login
password_changed_at	Datetime	Last password change
mfa_enabled	Boolean	MFA status
created_at	Datetime	Account creation

19.4 System Roles

Table Name: system_roles

Field	Data Type	Description
role_id	Integer	Unique role
role_name	Varchar	Role name
role_description	Text	Role description

Field	Data Type	Description
risk_level	Varchar	Low, medium, high, privileged
status	Varchar	Active or inactive

20. Information Security Tables

20.1 Security Incidents

Table Name: security_incidents

Field	Data Type	Description
security_incident_id	Integer	Unique incident
incident_number	Varchar	Incident reference
detected_date	Datetime	Detection date
reported_by	Integer	Reporter
incident_type	Varchar	Malware, phishing, access, fraud, data loss
affected_asset_id	Integer	Affected asset
branch_id	Integer	Branch
severity	Varchar	Low, medium, high, critical
incident_description	Text	Details
incident_status	Varchar	Open, contained, investigating, resolved
assigned_to	Integer	Incident handler
root_cause	Text	Root cause
resolution_date	Date	Resolution
financial_impact	Decimal	Estimated impact

20.2 Security Alerts

Table Name: security_alerts

Field	Data Type	Description
alert_id	Integer	Unique alert
alert_date	Datetime	Alert time
alert_source	Varchar	SIEM, firewall, antivirus, application

Field	Data Type	Description
alert_type	Varchar	Alert classification
user_id	Integer	Related user
asset_id	Integer	Related asset
source_ip	Varchar	Source IP
destination_ip	Varchar	Destination IP
severity	Varchar	Severity
alert_status	Varchar	New, investigating, closed, false positive
assigned_to	Integer	Analyst
resolution	Text	Resolution

20.3 Access Reviews

Table Name: access_reviews

Field	Data Type	Description
access_review_id	Integer	Unique review
review_period	Varchar	Review period
user_id	Integer	User reviewed
role_id	Integer	Role
reviewer_id	Integer	Reviewer
access_required	Boolean	Whether access remains required
review_decision	Varchar	Retain, modify, remove
review_date	Date	Date
action_status	Varchar	Pending, completed

20.4 Risk Register

Table Name: risk_register

Field	Data Type	Description
risk_id	Integer	Unique risk
risk_number	Varchar	Risk reference
risk_category	Varchar	Strategic, operational, financial, cyber
department_id	Integer	Owning department

Field	Data Type	Description
risk_description	Text	Risk statement
likelihood	Integer	Likelihood rating
impact	Integer	Impact rating
inherent_risk_score	Integer	Inherent score
control_description	Text	Existing controls
residual_risk_score	Integer	Residual score
treatment_plan	Text	Planned action
risk_owner_id	Integer	Risk owner
target_date	Date	Treatment date
risk_status	Varchar	Open, treating, accepted, closed

20.5 Audit Findings

Table Name: audit_findings

Field	Data Type	Description
finding_id	Integer	Unique finding
audit_number	Varchar	Audit reference
audit_date	Date	Audit date
department_id	Integer	Audited department
finding_type	Varchar	Conformity, observation, minor, major
finding_description	Text	Finding details
risk_rating	Varchar	Low, medium, high, critical
root_cause	Text	Root cause
corrective_action	Text	Corrective action
action_owner_id	Integer	Responsible employee
due_date	Date	Due date
finding_status	Varchar	Open, in progress, closed
closure_date	Date	Closure date

21. Business Intelligence Tables

21.1 KPI Definitions

Table Name: kpi_definitions

Field	Data Type	Description
kpi_id	Integer	Unique KPI
kpi_code	Varchar	KPI code
kpi_name	Varchar	KPI name
department_id	Integer	Responsible department
description	Text	KPI description
calculation_formula	Text	Calculation method
frequency	Varchar	Daily, weekly, monthly, quarterly
target_value	Decimal	Target
direction	Varchar	Higher is better or lower is better
owner_id	Integer	KPI owner
status	Varchar	Active or inactive

21.2 KPI Results

Table Name: kpi_results

Field	Data Type	Description
kpi_result_id	Integer	Unique KPI result
kpi_id	Integer	KPI
period_start	Date	Start
period_end	Date	End
branch_id	Integer	Branch
department_id	Integer	Department
actual_value	Decimal	Actual result
target_value	Decimal	Target
variance	Decimal	Difference
performance_status	Varchar	Below, meets, exceeds
calculated_at	Datetime	Calculation time

21.3 Dashboard Access Logs

Table Name: dashboard_access_logs

Field	Data Type	Description
dashboard_access_id	Integer	Unique log
dashboard_name	Varchar	Dashboard
user_id	Integer	User
access_date	Datetime	Access time
device_type	Varchar	Device
branch_id	Integer	Branch
access_status	Varchar	Successful or denied

22. Recommended Initial Database Scope

The first training database should begin with approximately 25 core tables.

Phase 1 Core Tables

1. regions
2. branches
3. departments
4. job_positions
5. employees
6. employee_attendance
7. payroll
8. customers
9. product_categories
10. products
11. suppliers
12. warehouses
13. inventory_balance
14. inventory_transactions
15. sales_orders
16. sales_order_items
17. customer_payments
18. purchase_orders
19. purchase_order_items
20. marketing_campaigns
21. customer_complaints
22. expenses
23. budgets
24. sales_targets

25. kpi_results

These tables are sufficient for introductory and intermediate training.

23. Database Relationship Overview

Key relationships include:

- One region has many branches.
 - One branch has many employees.
 - One department has many employees.
 - One job position has many employees.
 - One employee may manage many employees.
 - One customer has many sales orders.
 - One sales order has many sales order items.
 - One product appears in many sales order items.
 - One supplier supplies many products.
 - One supplier has many purchase orders.
 - One purchase order has many purchase order items.
 - One warehouse stores many products.
 - One product can exist in many warehouses.
 - One sales order can have many payments.
 - One sales order may have one or more deliveries.
 - One customer may have many complaints.
 - One employee may manage many complaints or support tickets.
 - One department may have many budgets, expenses, risks, and KPIs.
-

24. Primary Keys and Foreign Keys

Every table must have a unique primary key.

Examples:

- employee_id.
- customer_id.
- product_id.
- sales_order_id.
- branch_id.
- supplier_id.

Foreign keys connect tables.

Examples:

- employees.branch_id links to branches.branch_id.
- employees.department_id links to departments.department_id.
- sales_orders.customer_id links to customers.customer_id.

- sales_order_items.product_id links to products.product_id.
 - purchase_orders.supplier_id links to suppliers.supplier_id.
 - inventory_balance.warehouse_id links to warehouses.warehouse_id.
-

25. Data Quality Rules

The simulation database shall include both clean and intentionally messy versions.

25.1 Clean Database Rules

- Primary keys must be unique.
- Mandatory fields must not be blank.
- Email addresses must follow valid formats.
- Phone numbers must use approved formats.
- Dates must use consistent formats.
- Monetary values must not contain text.
- Foreign keys must reference valid records.
- Quantities must not be negative unless the transaction type permits it.
- Sales totals must match line items.
- Payroll totals must follow approved calculations.
- Duplicate master records must be prevented.

25.2 Messy Dataset Scenarios

Students will encounter:

- Duplicate customers.
- Missing product prices.
- Incorrect dates.
- Inconsistent gender values.
- Misspelled city names.
- Extra spaces.
- Mixed capitalization.
- Invalid emails.
- Negative quantities.
- Missing employee departments.
- Incorrect branch codes.
- Duplicate invoices.
- Inconsistent product categories.
- Blank salary values.
- Incorrect sales totals.
- Invalid phone numbers.

These issues will support data-cleaning lessons.

26. Data Classification

ABC Retail Ltd simulation data should be classified as follows:

Classification	Examples
Public	Product catalog, branch names
Internal	Sales targets, inventory levels
Confidential	Payroll, customer information
Restricted	Passwords, security incidents, bank details

All training data must be fictional and must not contain real customer, employee, banking, or confidential company information.

27. Data Privacy Rules

The simulation should use:

- Fictional names.
- Fictional addresses.
- Fictional phone numbers.
- Fictional email addresses.
- Masked bank information.
- Synthetic financial transactions.
- Synthetic security incidents.
- Synthetic medical or sensitive fields only where absolutely necessary.

Students must be taught not to use real personal data in public portfolios.

28. Excel Training Use Cases

The enterprise model will support Excel lessons such as:

- Data entry.
- Formatting.
- Removing duplicates.
- Handling blanks.
- Text functions.
- Date functions.
- IF statements.
- SUMIFS.
- COUNTIFS.
- XLOOKUP.
- Pivot tables.

- Charts.
 - Data validation.
 - Conditional formatting.
 - Dashboard development.
 - Budget variance analysis.
 - Payroll analysis.
 - Sales reporting.
 - Inventory monitoring.
-

29. SQL Training Use Cases

Students will practice:

- SELECT.
- WHERE.
- ORDER BY.
- DISTINCT.
- GROUP BY.
- HAVING.
- JOIN.
- UNION.
- CASE.
- Subqueries.
- Common table expressions.
- Views.
- Stored procedures.
- Window functions.
- Database normalization.
- Data integrity.
- Indexing.
- Role-based access.

Example questions include:

- Which branch generated the highest revenue?
 - Which products are below reorder level?
 - Which customers have overdue balances?
 - Which employees exceeded sales targets?
 - Which suppliers delivered late?
 - Which marketing campaign produced the most customers?
 - Which departments exceeded their budgets?
 - Which branches have high customer complaint rates?
-

30. Power BI Training Use Cases

Students will build dashboards for:

- Executive performance.
- Sales.
- Finance.
- Human resources.
- Inventory.
- Procurement.
- Customer service.
- Marketing.
- Logistics.
- Information security.
- Risk management.

Students will practice:

- Data import.
- Power Query.
- Data modeling.
- Relationships.
- DAX.
- Measures.
- Calculated columns.
- Date tables.
- Drill-through.
- Tooltips.
- Filters.
- Row-level security.
- Dashboard design.
- Storytelling.

31. Artificial Intelligence Training Use Cases

AI projects may include:

- Sales forecasting.
- Customer segmentation.
- Product recommendations.
- Customer churn prediction.
- Inventory demand forecasting.
- Complaint classification.
- Employee attrition prediction.
- Fraud detection.
- Supplier risk scoring.
- Security alert classification.
- Automated management summaries.

- Natural-language querying of business data.
-

32. Capstone Project Structure

The final capstone project may require students to act as analysts hired by ABC Retail Ltd.

The student team receives:

- Business background.
- Management questions.
- Raw datasets.
- Data dictionary.
- Data-quality problems.
- Departmental targets.
- Financial data.
- Sales data.
- Customer data.
- Inventory data.
- Operational risks.

Students must:

1. Understand the business problem.
 2. Clean the data.
 3. Document data-quality issues.
 4. Build the data model.
 5. Conduct analysis.
 6. Design a dashboard.
 7. Produce insights.
 8. Make recommendations.
 9. Present findings.
 10. Defend the project.
-

33. Capstone Business Questions

Possible questions include:

- Why did revenue decline in some branches?
- Which products are underperforming?
- Which customers contribute the most revenue?
- Which regions have the highest growth potential?
- Which departments exceed budgets?
- Which suppliers create operational delays?
- Which products are likely to run out of stock?
- Which branches have high employee absenteeism?
- Which campaigns generate the best return?
- What are the main causes of customer complaints?

- Which security risks require immediate action?
-

34. Data Release Levels

Level 1 – Beginner

Small Excel files containing:

- 50 to 500 rows.
- Few columns.
- Basic errors.
- Simple formulas.

Level 2 – Intermediate

Connected files containing:

- 1,000 to 20,000 rows.
- Multiple tables.
- More complex cleaning.
- Pivot tables and joins.

Level 3 – Advanced

Relational databases containing:

- 50,000 to 500,000 records.
- Multiple domains.
- Historical transactions.
- Advanced SQL and Power BI models.

Level 4 – Professional Simulation

Enterprise-scale synthetic data containing:

- More than one million records.
 - Incremental refresh.
 - Security roles.
 - Forecasting.
 - Machine learning.
 - Executive dashboards.
 - Data governance controls.
-

35. Recommended Dataset Calendar

Week	Dataset
Week 1	Employees and Departments
Week 2	Customers and Products
Week 3	Sales and Sales Items
Week 4	Inventory and Warehouses
Week 5	Suppliers and Purchase Orders
Week 6	Finance and Expenses
Week 7	Marketing and Customer Service
Week 8	Integrated Executive Dataset

36. File Naming Standards

Recommended naming format:

ABC_Department_Dataset_Level_Version

Examples:

- ABC_Sales_Beginner_v1.xlsx
 - ABC_HR_Attendance_Intermediate_v1.xlsx
 - ABC_Inventory_Clean_v1.xlsx
 - ABC_Inventory_Messy_v1.xlsx
 - ABC_Enterprise_SQL_v1.sql
 - ABC_Executive_PowerBI_v1.pbix
-

37. Dataset Version Control

Every dataset should include:

- Dataset name.
- Version.
- Creation date.
- Owner.
- Description.
- Number of rows.
- Number of columns.
- Data classification.
- Intended lesson.

- Known issues.
- Change history.

Datasets should not be modified without updating the version.

38. Database Development Phases

Phase 1 – Foundation

Build 25 core tables.

Phase 2 – Departmental Expansion

Expand to 50 tables.

Phase 3 – Enterprise Integration

Expand to 75 tables.

Phase 4 – Governance and Security

Expand to 100 tables.

Phase 5 – Advanced Analytics

Add machine-learning datasets, historical snapshots, and analytical marts.

39. Expected Final Deliverables

The ABC Retail Ltd simulation package should eventually contain:

- Enterprise profile.
- Organization chart.
- Business processes.
- Data dictionary.
- Entity relationship diagram.
- SQL database.
- SQL scripts.
- Excel workbooks.
- Messy datasets.
- Clean datasets.
- Power BI model.
- DAX measures.
- Dashboards.
- Business scenarios.

- Student exercises.
 - Instructor solutions.
 - Assessment questions.
 - Capstone project.
 - Marking rubrics.
 - Data governance guide.
 - Security scenarios.
 - AI project datasets.
-

40. Approval

Prepared by:

Director of Academic Affairs

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Signature: ____

Date: ____

Reviewed by:

Director of Technology and Innovation

Name: ____

Signature: ____

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